C Programming Project Presentation

Student Attendance Management System (With Files)

KGiSL MicroCollege
Genius as a Service (GaaS)
TECHNICAL TEAM

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KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
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Objective:

The main aim of the Student Attendance Management System in C, utilizing file handling, is to digitize and automate the manual task of tracking and reporting student attendance. Instead of relying on physical registers, the program persists student records (including name, roll number, and accumulated attendance counts) by saving them to a binary data file on the disk. This allows the system to efficiently handle three core functions: adding new students, safely marking daily attendance (updating the running totals using a temporary file to maintain data integrity), and finally, generating clear reports that display each student's attendance percentage, providing administrators or instructors with a quick, data-driven overview of class performance.

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Module:

1. Main Module (main()):- Controls program execution and user selections.
2. Menu Module (menu()):- Displays available operations
3. Add Student Module (add()):- Inputs new student details and saves them to file
.4. Mark Attendance Module (mark()):- Updates student attendance status and saves changes
.5. View Records Module (view()):- Displays attendance details of all students.
6. File Handling Module:- Used internally to store and retrieve data in binary f

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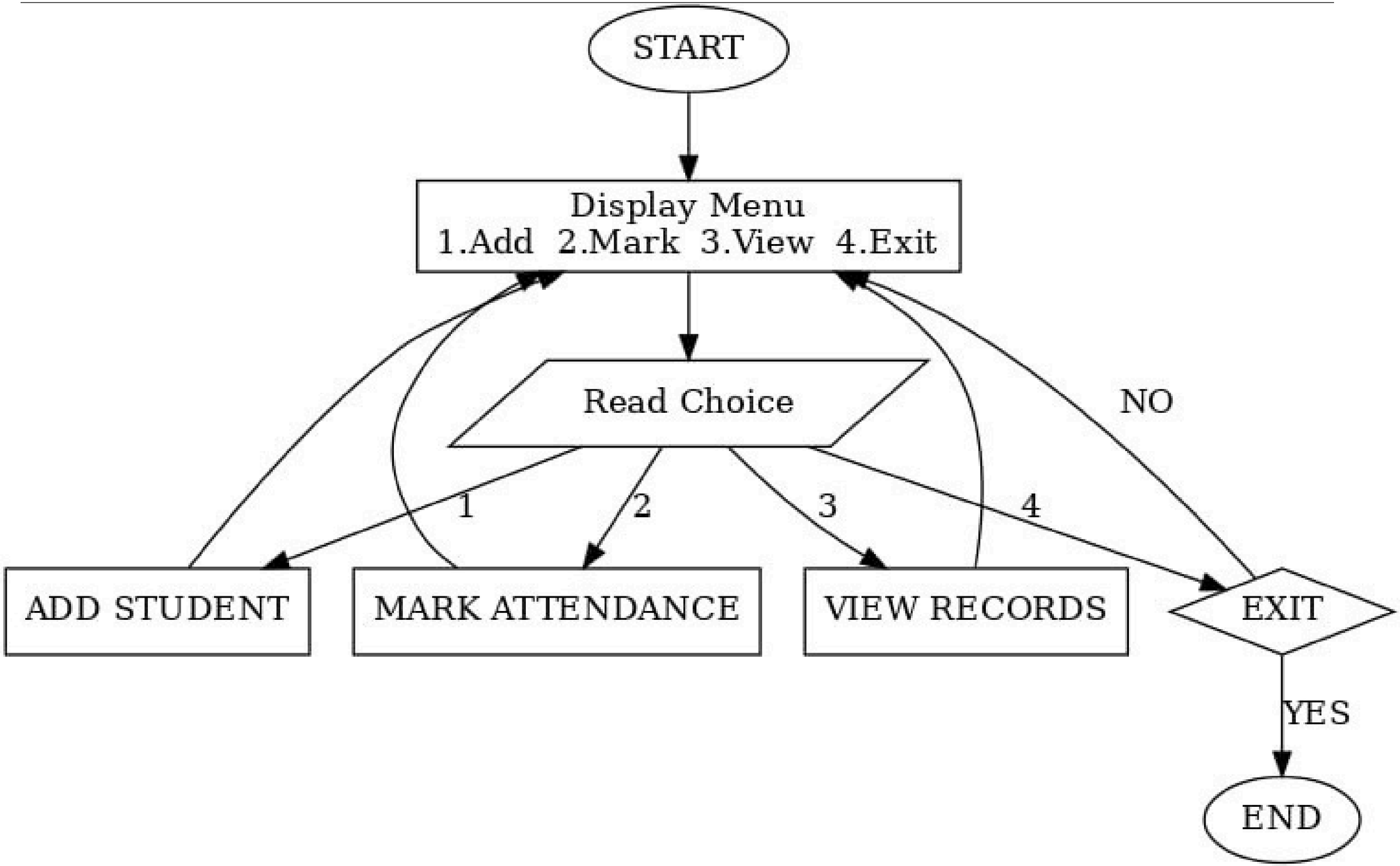

Program :

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define FILENAME "data.txt"

struct Student {
    int roll;
    char name[50];
    int present;
    int total;
};

void menu();
void add();
void mark();
void view();
```

```
int main() {
    int choice;

    do {
        menu();
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                add();
                break;
            case 2:
                mark();
                break;
            case 3:
                view();
                break;
```

```
            case 4:
                printf("\nBye!\n");
                break;
            default:
                printf("\nWrong choice, try
again.\n");
        }
    } while (choice != 4);

    return 0;
}

void menu() {
    printf("\n--- ATTENDANCE SYSTEM ---\n");
    printf("1. Add Student\n");
    printf("2. Mark Attendance\n");
    printf("3. View Records\n");
    printf("4. Exit\n");
    printf("Enter choice: ");
```

```
}

void add() {
    FILE *f;
    struct Student s;

    printf("\n--- Add New Student ---\n");

    printf("Enter Roll Number: ");
    scanf("%d", &s.roll);

    printf("Enter Name (one word): ");
    scanf("%s", s.name);

    s.total = 0;
    s.present = 0;

    f = fopen(FILENAME, "ab");
```

```
    if (f == NULL) {
        printf("\nERROR: Cannot open file.\n");
        return;
    }

    fwrite(&s, sizeof(struct Student), 1, f);
    fclose(f);

    printf("\nStudent %s added.\n", s.name);
}

void mark() {
    FILE *f;
    struct Student s;
    int target_roll;
    char status;
    int found = 0;
```

```
    printf("\n--- Mark Attendance ---\n");
    printf("Enter Roll to mark: ");
    scanf("%d", &target_roll);

    f = fopen(FILENAME, "r+b");
    if (f == NULL) {
        printf("\nFile error or no students added
yet.\n");
        return;
    }

    while (fread(&s, sizeof(struct Student), 1, f)
== 1) {
        if (s.roll == target_roll) {
            found = 1;

            printf("Is %s Present? (P/A): ",
s.name);
            scanf(" %c", &status);
```

```
            s.total++;
            if (status == 'P' || status == 'p') {
                s.present++;
                printf("Marked: PRESENT.\n");
            } else if (status == 'A' || status ==
'a') {
                printf("Marked: ABSENT.\n");
            } else {
                s.total--;
                printf("Invalid status. No update
done.\n");
            }
            fseek(f, -(long)sizeof(struct Student),
SEEK_CUR);

            fwrite(&s, sizeof(struct Student), 1,
f);
```

```
            break;
        }
    }

    fclose(f);

    if (!found) {
        printf("\nRoll %d not found.\n",
target_roll);
    } else {
        printf("Record updated.\n");
    }
}

void view() {
    FILE *f;
    struct Student s;
    int count = 0;
```

```
    f = fopen(FILENAME, "rb");
    if (f == NULL) {
        printf("\nNo data file found.\n");
        return;
    }

    printf("\n--- All Records ---\n");
    printf("Roll | Name | Present | Total\n");
    printf("----------------------------\n");

    while (fread(&s, sizeof(struct Student), 1, f)
== 1) {
        count++;
        printf("%d | %s | %d | %d\n",
               s.roll,
               s.name,
               s.present,
               s.total);
    }
```

```
    printf("-------------------------------\n");

    if (count == 0) {
        printf("No students saved yet.\n");
    }

    fclose(f);
}
```

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Output

```
C:\Users\KGCAS\Documents\I

--- ATTENDANCE SYSTEM ---
1. Add Student
2. Mark Attendance
3. View Records
4. Exit
Enter choice: 1

--- Add New Student ---
Enter Roll Number: 123
Enter Name (one word): srinesh

Student srinesh added.

--- ATTENDANCE SYSTEM ---
1. Add Student
2. Mark Attendance
3. View Records
4. Exit
Enter choice: 2

--- Mark Attendance ---
Enter Roll to mark: 123
Is srinesh Present? (P/A): p
Marked: PRESENT.
Record updated.

--- ATTENDANCE SYSTEM ---
1. Add Student
2. Mark Attendance
3. View Records
4. Exit
Enter choice: 3

--- All Records ---
Roll | Name | Present | Total
----------------------------
123 | srinesh | 2 | 2
123 | srinesh | 0 | 0
----------------------------
```

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```
Record updated.

--- ATTENDANCE SYSTEM ---
1. Add Student
2. Mark Attendance
3. View Records
4. Exit
Enter choice: 3

--- All Records ---
Roll | Name | Present | Total
-----------------------------
123 | srinesh | 2 | 2
123 | srinesh | 0 | 0
-----------------------------

--- ATTENDANCE SYSTEM ---
1. Add Student
2. Mark Attendance
3. View Records
4. Exit
Enter choice: 4

Bye!

--------------------------------
Process exited after 85.27 seconds with return value 0
Press any key to continue . . .
```

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Conclusion

This Attendance Management System helps to store and manage student attendance in an easy and organized way. It allows adding new student records, marking attendance for each day, and viewing the attendance details whenever needed. The use of file handling ensures that the data is saved permanently and can be accessed later. This program is simple to use and reduces manual work, making attendance maintenance faster, accurate, and efficient.

Thank You